

CURRICULUM VITAE

Jamshid Derakhshan

6 July 2026

PERSONAL INFORMATION.

Born: May 1975, Oxford, UK

Married

Two daughters (born in 2017, 2021)

Address:

St Hilda's College, Cowley Place, Oxford OX4 1DY

St Peter's College, New Inn Hall St, Oxford OX1 2DL

Email: derakhsh@maths.ox.ac.uk

<https://www.st-hildas.ox.ac.uk/people/jamshid-derakhshan>

<https://www.spc.ox.ac.uk/people/jamshid-derakhshan>

EMPLOYMENT.

- **Auxiliary Dean of Degrees**, St Hilda's College, since 06/26.
- **Lecturer in Pure Mathematics**, St Peter's College, University of Oxford, since 10/2024.
- **Lecturer in Pure Mathematics**, St Hilda's College, University of Oxford, since 10/2014 (Permanent appointment).
- **Dean of Degrees**, St Hilda's College, University of Oxford, 11/2020-7/2023.
- **Deputy Dean of Degrees**, St Hilda's College, University of Oxford, 9/2016-11/2020.
- **Member**, Institute des Hautes Etudes Scientifiques (IHES), Bures-sur-Yvette, France, 3/2018-4/2018.
- **Departmental Lecturer in Pure Mathematics**, Mathematical Institute, University of Oxford, 10/2013 - 9/2014.
- **Academic Visitor**, Mathematical Institute, University of Oxford, 10/2004-9/2013.
- **Member**, Max Planck Institute for Mathematics (MPIM), Bonn, Germany, 5/2012.
- **Visitor**, Center for Advanced Studies in Mathematics, Ben Gurion University of the Negev, Beer Sheva, Israel, 12/2010-3/2011.

- **Member**, Institute des Hautes Etudes Scientifiques (IHES), Bures-sur-Yvette, France, 10/2002-12/2002.
- **Visiting Researcher**, Institute of Mathematics, Hebrew University of Jerusalem, Jerusalem, Israel, 10/2002-9/2004.
- **Postdoctoral Fellow**, Emmy Noether Institute, Bar Ilan University, Israel, 10/1999-9/2002.
- **Golda Meir Postdoctoral Fellow**, Institute of Mathematics, Hebrew University of Jerusalem, Jerusalem, Israel, 10/1997-9/1999.
- **Lady Davis Postdoctoral Fellow**, Institute of Mathematics, Hebrew University of Jerusalem, Jerusalem, Israel, 10/1997-9/1999.

EDUCATION.

- **D.Phil (PhD) in Mathematics**, 21/8/1996,
Merton College, University of Oxford.
Supervisor: Prof. Frank Wagner (1993-1996), Prof. Angus Macintyre FRS (1992-1993)
Thesis title: *Problems on nilpotency and local finiteness in infinite groups and infinite dimensional algebras*

D.Phil obtained at age 21. Second youngest D.Phil in Mathematics at Oxford University history after Ruth Lawrence.

- **I was self-taught in mathematics from age 12 and proved new results in descriptive set theory, group theory, and model theory from age 13** (while I was at school in Tehran Iran) which I wrote in 13 manuscripts until the age of 16 when I was admitted to Oxford University because of my research works. I went to Oxford at age 17 and directly became a D.Phil student.

Oxford University Prizes, Scholarships, and Awards:

- **Senior Mathematical Prize 1995**, University of Oxford, July 1995, for D.Phil Thesis.
- Youngest ever recipient at age 20 of the Senior Mathematical Prize of Oxford University.
- **Johnson University Prize 1995**, University of Oxford, July 1995, for the "Doctorate thesis of the greatest merit" at Oxford University.
- Youngest ever recipient at age 20 of Oxford University's Johnson University Prize.

(Senior Mathematical Prize and Johnson Prize Certificates were signed by Professors Simon Donaldson, Ion James, Roger Penrose, and Daniel Quillen).

- **Senior Scholarship** at Merton College, University of Oxford, 1993-1996. (Full scholarship).
- Youngest ever Senior Scholar at Merton College, Oxford, appointed at age 17.
- **ORS Award**, UK Government, 1993-1996.
- **Book Prize 1996** from Warden of Merton College, July 1996.

- **Commendation** from Mathematical Institute, University of Oxford, July 1994.

FUNDS, HONOURS.

- **John O’Conner Research Fund June 2026, St Peter’s College.** Annual Research Grant Awarded by St Peter’s College, given to me for visiting Prof. Angus Macintyre at Edinburgh and Prof. Laurent Lafforgue in Paris.

- **Talk by Prof. Ehud Hrushovski on our work.**

Professor Ehud Hrushovski (Oxford) gave a talk in the conference on "Model theory and groups on the occasion of Katrin Tent’s 60th Birthday" in Muenster, 25 - 29 September 2023, on our joint work. Title of his talk: "*From p -adic to adelic imaginaries, via Borel equivalence relations*".

- **Talk by Prof. Thomas Scanlon on my work with Macintyre solving a question of Ax that was open from 1968.**

My solution with Angus Macintyre to a problem of James Ax posed in 1968 on the decidability of the class of all the rings $\mathbb{Z}/m\mathbb{Z}$, $m > 1$ was stated by Professor Thomas Scanlon (Berkeley) in his talk at MSRI Berkeley on 12 October 2021.

- **My works on uniform definition for valuation rings and model completeness for finite extensions of p -adics were stated in Jochen Koenigsmann’s talk in International Congress of Mathematicians ICM 2018 in Rio**

My work with Raf Cluckers, Eva Leenknegt, and Angus Macintyre on uniform definability in ring language of the the valuation rings of all Henselian valued fields with finite or pseudo-finite residue fields (Ann. Pure and Applied Logic, 2013) and my work with Angus Macintyre on model completeness in the ring language for finitely ramified Henselian fields (Confluentes Math 2023) were stated in Jochen Koenigsmann’s talk in the International Congress of Mathematicians ICM 2018 in Rio (cf. Proc. ICM 2018, World Scientific Publishing company, p. 68-onwards).

- **Visiting and Adjunct Professorship, Institute for Research in Fundamental Sciences (IPM), Tehran,** two months 12/2017 and 9/2018.
- **Member Max Planck Institute for Mathematics (MPIM), Bonn,** Semester in Model Theory and Applications, 6/2012.
- **Member Institute des Hautes Etudes Scientifiques (IHES),** in 12/2002 and 4/2018.
- **Fellowship award, Center for Advanced Studies in Mathematics, Ben Gurion University, Israel,** January – March 2010.

- **Postdoctoral Fellowship, and Funds from US-Israel Bi-National Science Foundation and Marie Curie funds from EU, Department of Mathematics and Emmy Noether Mathematics Institute, Bar Ilan University, Israel, 1999.**
- **Golda Meir Fellowship, Institute of Mathematics, Hebrew University of Jerusalem, Israel, 1998.**
- **Lady Davis Fellowship, Institute of Mathematics, Hebrew University of Jerusalem, Israel, 1997.**

RESEARCH INTERESTS.

Mathematical Logic, Model Theory and its connections to Algebraic Geometry, Group Theory, and Number Theory.

More specifically:

- Algebraic groups and Abelian varieties over local fields, global fields, and adeles
- p -adic and motivic integration, and applications to algebra and number theory.
- Model theory of adeles.
- Model theory of sheaves.
- Model theory of valued fields.
- Geometric stability theory.
- Model theory of groups and rings.
- Hodge theory via o-minimality and adelic analogs.
- Zeta and L -functions and related topics in number theory and Diophantine geometry.
- Non-Abelian adelic Fourier transform, and the Langlands conjectures.
- Conjectures of numbers of rational points on varieties and homogeneous spaces.
- Model theory, cohomology, algebraic geometry.
- Measures and analysis on adelic spaces.
- Representation theory over p -adic fields and over adeles.

WORK COVERED IN UNIVERSITY OF OXFORD NEWS.

An article on my solution with Angus Macintyre of a problem of Ax asked in 1968 was published in St Hilda's College News:

<https://www.st-hildas.ox.ac.uk/content/fifty-year-old-maths-problem-finally-solved-st-hildas> (**Fifty year old maths problem finally solved at St Hilda's**). Reference for our paper: Forum of Mathematics, Sigma, Vol 11, July 2023.

PROFESSIONAL ACTIVITIES.

- . **Editorial Responsibilities**

- Editor of *Book Series: World Scientific Studies in Pure Mathematics*, World Scientific Publishing company (other editors: Jacob Tsimerman, Yakov Kremnitzer)

- Editor of the series *Landscapes*, College Publications, London

. Committees, advisory boards, examinations

- Member of panel for evaluating EPSRC small grants applications, October 2025.

- Member of interviewing team and evaluating and admitting Oxford undergraduate students in Mathematics and joint schools at St Hilda's College, Oxford, every year since 2014.

- Member of the Jury for Habilitation Thesis Defense of Dr Olivia Caramello, *Grothendieck toposes as unifying bridges in mathematics*, University of Paris 7. 14/12/2016. Chairman of Jury was Prof. Alain Connes (Fields Medalist).

- Internal examiner for the D.Phil thesis of Wojciech Wołoszyn (Supervisors: Ehud Hrushovski, Joel David Hamkins, W. Hugh Woodin), June 2026.

- Internal examiner for the D.Phil thesis of Benedikt Stock (Supervisor: Jochen Koenigsmann), January 2026.

- Internal Examiner for transfer of status PRS to D.Phil, Mira Tartarotti (Supervisor: Ehud Hrushovski), 10 Oct. 2025.

- Internal Examiner for the D.Phil Thesis of Konstantinos Kartas (Supervisors: Ehud Hrushovski, Jochen Koenigsmann), on *Contributions to the model theory of Henselian fields*, University of Oxford, Viva on 21/6/2022.

- Internal Examiner for the D.Phil Thesis of Benjamin Rigler (Supervisor: Jochen Koenigsmann), on *Decidability in extensions of $F_p((t))$* , University of Oxford, Viva on 10/2020.

- Internal Examiner for the D.Phil Thesis of Victor Lisinski (Supervisor: Jochen Koenigsmann and Ehud Hrushovski), on *Model theory and algebra of positive characteristic Hahn fields*. University of Oxford, Viva on 15 January 2022.

- Internal Examiner for the MSc by Research Thesis of Chenkai Wang (Supervisor: Jochen Koenigsmann), *On the decidability of $\mathbb{Q}_p^{ab}$* , University of Oxford, Viva on 11/2018.

. Organisation of Conferences and Seminars

- Local Organising Committee for *Geometry from the model theorist's point of view: A conference in honour of Boris Zilber's 75th birthday*, 10-13 September 2024, Mathematical Institute and St Hilda's College, Oxford.

- Organizer of the Mathematical Logic Seminar, Hebrew University of Jerusalem, Israel, during 1998-1999.

PUBLICATIONS.

16. J. Derakhshan, E. Hrushovski, *Imaginaries, products, and the adèle rings*, **Volume for Katrin Tent**, EMS, to appear. (arXiv: 2309.11678)

15. J. Derakhshan, A. Macintyre, *Model theory of Abelian groups that are finite-by-Presburger, and applications to p -adic fields*, **Contemporary Model Theory**, Handbook Edited by A. S. Daghighi, J. Derakhshan, M. Pourmahdian, Landscapes, College Publications, London, to appear. (arXiv: 1603.08601).

14. J. Derakhshan, *Model theory of adèles and number theory*, **Contemporary Model Theory**, Handbook Edited by A. S. Daghighi, J. Derakhshan, M. Pourmahdian, Landscapes, College Publications, London, to appear. (arXiv: 2007.09237).

13. J. Derakhshan, *p -adic Model theory, p -adic integrals, Euler products, and zeta functions of groups*, **Contemporary Model Theory**, Handbook Edited by A. S. Daghighi, J. Derakhshan, M. Pourmahdian, Landscapes, College Publications, London, to appear. (On arXiv: 2007.09242).

12. J. Derakhshan, A. Macintyre, *Axioms for commutative unital rings elementarily equivalent to restricted products of connected rings*, **Notre Dame Journal of Formal Logic**, 67 (1), 1-14 (2026). (arXiv: 2007.09244).

11. J. Derakhshan, A. Macintyre, *Model Completeness for Henselian fields with finite ramification valued in a $\mathbb{Z}$ -group*, **Confluentes Mathematici**, 15, (2023) 137-147.

10. J. Derakhshan, A. Macintyre, *Decidability of all the rings $\mathbb{Z}/m\mathbb{Z}$: A problem of Ax*, **Forum of Mathematics, Sigma**, 11 (2023) e64(1-6).

9. J. Derakhshan, A. Macintyre, *Model theory of adèles I*, **Annals of Pure and Applied Logic**, 173, 3, March 2022. (arXiv: 1603.09698).

8. P. D'Aquino, J. Derakhshan, A. Macintyre, *Truncations of ordered abelian groups*, **Algebra Universalis**, 82 (2021), no. 2. (arXiv: 1910.14517).

7. J. Derakhshan, A. Macintyre, *Enrichments of Boolean algebras by Presburger predicates*, **Fundamenta Mathematicae**, issue 1, volume 239, (2017), 1-17.

6. J. Derakhshan, *Model theory and zeta functions*, **Mathematisches Forschungsinstitut Oberwolfach**, Report No. 49/2016 (Definability and Decidability Problems in Number Theory).

5. J. Derakhshan, A. Macintyre, *Some supplements to Feferman-Vaught related to the model theory of adeles*, **Annals of Pure and Applied Logic**, 165 (11) (2014), 1639-1679.

4. R. Cluckers, J. Derakhshan, E. Leenknegt, A. Macintyre, *Uniformly defining valuation rings in Henselian valued fields with finite or pseudo-finite residue field*, **Annals of Pure and Applied Logic**, 164 (2013), 1236-1246.

3. M. Berman, J. Derakhshan, P. Paaanen, U. Onn, *Uniform cell decomposition with applications to Chevalley groups*, **Journal of the London Mathematical Society**, 87(2) (2013), 586-606.

2. J. Derakhshan, F. Wagner *Skew linear unipotent groups*, **Bulletin of the London Mathematical Society**, 38 (2006), No. 3, 44 -449.

1. J. Derakhshan, F. Wagner *Nilpotency in groups with chain conditions*, **Quarterly Journal of Mathematics Oxford**, Ser. (2), 48 (1997), No. 192, 453-466.

PREPRINTS IN PREPARATION.

1. J. Derakhshan, *Definable sets, Euler products of p -adic integrals, and zeta functions I: meromorphic continuation*, Preprint under expansion.

2. J. Derakhshan, *Definable subgroups of algebraic groups over non-archimedean local fields*, Preprint under expansion.

3. J. Derakhshan, A. Macintyre, *Decidability problems for adèle rings and related restricted products*, **arXiv: 1910.14471**, (We are extending the results of this paper). This paper has been re-titled to: *Model Theory of Adeles II: elementary equivalence and uniform decidability*.

4. J. Derakhshan, *Solubility in groups with chain condition on uniformly definable subgroups*, **Archiv for Mathematical Logic**, Under Revision. **arXiv: 1911.01589**.

5. J. Derakhshan, A. Macintyre, *Model theory of adeles III: Adelic measures and integrals*.

6. J. Derakhshan, R. Takloo-Bighash, *Definable adelic integrals*. [This is start of our collaboration connecting model theory of adeles to representation theory of automorphic forms and so may be incorporated later into future works].

MONOGRAPHS IN PREPARATION.

1. J. Derakhshan, L. Lafforgue, **Langlands Functoriality, Nonabelian Fourier Transform, and Poisson Formulas**, World Scientific Publishing Company, contract signed on 9 June 2022.

(In this book we prove that the Functoriality Conjecture of Langlands is equivalent to a non-abelian adelic Poisson summation formula generalizing the Poisson summation formula for ideles due to Tate. This formula holds true over function fields).

2. J. Derakhshan, **Model theory of Adeles and Number Theory**, World Scientific Publishing Company, contract signed on 18 January 2022.

(This book covers model-theoretic results on adeles and connections to number theory in joint works with Angus Macintyre and with Ehud Hrushovski).

3. J. Derakhshan, **Zeta Functions, p -adic and Motivic Integrals and Model Theory**, World Scientific Publishing Company, contract signed September 2016.

(This book presents my work on analytic properties and analytic continuation of Euler products of definable p -adic integrals and connections to problems on counting subgroups and representations, and in number theory counting rational points of bounded height).

4. C. de'Elbée, J. Derakhshan, **From nil rings to periodic groups: structure of groups and algebras satisfying identities**. World Scientific publishing company, contract just signed.

(This book presents results in ring theory related to nilotency of nil rings (Kurosh-Levisky, Shirshov) and group theory including works of Zelmanov around the Burnside problem and relates results for pro- p groups and Golod-Shafarevich condition.

TEXTBOOKS IN PREPARATION.

1. J. Derakhshan, **Topology**, Word Scientific Publishing Company, contract signed on 21 September 2017. (A first course in Topology - equivalent to Oxford's 2nd year course).

2. J. Derakhshan, **Rings and Modules**, World Scientific Publishing Company, contract signed on 21 September 2017. (A first course in Rings and Modules - equivalent to Oxford's 2nd year course).

EDITED VOLUMES.

1. J. Derakhshan, H. Furstenberg, *Fields Medalist's Lectures*, 5th edition, World Scientific Publishing Company, contract signed on 15 August 2022. [**The publisher is not going ahead with this at the moment, they may do later**]

2. J. Derakhshan, M. Pourmahdian, A. Sadegh Daghighi, *Contemporary Model Theory*, College Publications, London, to appear.

LECTURE COURSE TEACHING AT MATHEMATICAL INSTITUTE, UNIVERSITY OF OXFORD.

- C.1.1. Model Theory, 4th year/Master's Course, Mathematical Institute, University of Oxford, Michaelmas Term 2025.
- B3b. Algebraic Curves, 3rd year undergraduate course, Hilary Term 2014.

TUTORIAL TEACHING AT ST HILDA'S COLLEGE AND ST PETER'S COLLEGE (EVERY YEAR).

Introduction to University Mathematics (1st year)
 Prelims Analysis I, II, III (1st year)
 Linear Algebra I (1st year)
 Linear Algebra II (1st year)
 Groups and Group Actions (1st year)
 Constructive Mathematics (1st year)
 Linear Algebra (2nd year)
 Topology (2nd year)
 Rings and Modules (2nd year)
 Number Theory (2nd year)
 Projective Geometry (2nd year)
 Introduction to Manifolds and Multivariate Calculus (2nd year)
 Metric Spaces (2nd year)
 Complex Analysis (2nd year)

INTER-COLLEGIATE CLASSES AT THE MATHEMATICAL INSTITUTE, OXFORD.

Algebraic Curves (3rd year)
 Logic (3rd year)
 Set Theory (3rd year)
 Galois Theory (3rd year)
 Algebraic Number Theory (3rd year)
 Commutative Algebra (4th year)
 Algebraic Geometry (4th year)
 Lie Algebras (4th year)
 Lie Groups (4th year)
 Analytic Number Theory (4th year)
 Elliptic Curves (4th year)
 Model Theory (4th year)

ADVANCED COURSES.

- *Interactions of Model Theory and Number Theory: Adeles, Euler Products, Counting Rational Points and the Langlands Program*, Institute for Research in Fundamental Sciences (IPM), Tehran (Academic year 2017-2018).
- *Model theory of adeles and number theory*, Mathematical Institute, University of Oxford, (Academic year 2016-2017).
- *Model theory of finite and pseudo-finite fields*, Mathematical Institute, University of Oxford, (Academic year 2013-2014).
- *p-adic integration, cell decomposition, resolution of singularities, and rationality of Poincare series - after Denef, Loeser, Cluckers, Hrushovski, Kazhdan*, University of Oxford, (Academic year 2011-2012).

SOME INVITED TALKS.

- Model Theory of adeles, **A85: Algebra and Arithmetic from Model Theory** (Workshop on honour of Angus Macintyre's 85th birthday), **International Centre for Mathematical Sciences (ICMS), Edinburg**, 11-15 May 2026.
- Imaginaries in products and the ring of adeles (joint work with Ehud Hrushovski), 33rd meeting of **Lancashire Yorkshire Model Theory Seminar**, 23 January 2025, University of Manchester.
- **Model Theory and Applications to Groups and Combinatorics, CIRM, Luminy**, *Imaginaries in products and the ring of adeles (joint work with Ehud Hrushovski)*, 2 October 2024.
- **Oberseminar Algebra und Geometrie, Heinrich-Heine Universitat, Dusseldorf**, *Model theory of adeles*, 19 July 2024
- Logic Advanced Class, **Mathematical Institute, Oxford** *Model theory of Booleanizations, products, and sheaves of structures*, 7 March 2024.
- Online Logic Seminar, **University of Southern Illinois**, *Decidability of the class of all $\mathbb{Z}/m\mathbb{Z}$: A problem of Ax*, 8 February 2024.
- Logic Seminar, **Mathematical Institute, Oxford University**, *Imaginaries in products and in adèle rings (joint work with Ehud Hrushovski)*, forthcoming, Nov 2023.
- Logic Seminar, **Mathematical Institute, University of Oxford**, *Decidability of the class of all the rings $\mathbb{Z}/m\mathbb{Z}$: A problem of Ax*, 26/1/2023.
- **Institute for Research in Fundamental Sciences (IPM), Tehran, Iran**
 - Mathematical Logic Seminars, 10/2020, (2 Talks)
Video of talks: <http://math.ipm.ac.ir/logic/videos.jsp>
 - Mathematics Colloquium, 26/9/2018.
- **Frontiers in Mathematical Sciences, 5th Conference**
Institute for Research in Fundamental Sciences (IPM), Tehran, Iran, 11-13 July 2017.
- **Model Theory Conference, Bedlewo, Poland**
2-8 July 2017.
Video of talk: <https://www.youtube.com/watch?v=z4oc7s1RGr4>
- **MFO, Obervolfach**
Decidability and Definability in Number Theory, 10/2017.

- Logic, Zeta Functions, and Representations, **SEEMOD, Oxford**, July 5-6, 2016.
- **University of Manchester**
Logic Seminar, November 2016.
- **University of Oxford**
Logic Seminar, May 2016.
- **International Center for Mathematical Sciences, Edinburgh**
Recent Developments in Model Theory and its applications in Algebraic, Analytic and Diophantine Geometry (Workshop in Honor of Angus Macintyre), July 2014.
- **University of Oxford**
Logic Seminar, 19 June 2014, 12 June 2014, 28/11/2013, 24/1/2013, 10/5/2012, 20/10/2011, 16/6/2011, 23/6/2011, 5/5/2011.
- **University of Leeds**
Logic Seminar, 28 May 2014.
- **Princeton University/Institute for Advanced Studies**
Joint Number Theory Seminar, Princeton University Mathematics department, 8 May 2014.
- **City University of New York Graduate Center, New York**
Algebra and Cryptography Seminar, 9 May 2014.
- **University of California, Los Angeles**
Logic Seminar, 23 May 2014.
- **University of California, San Diego**
Special Algebra Seminar, 19 May 2014.
- **University of Manchester**
Logic Seminar, 20/11/2013.
- **University of Nottingham**
Number Theory Seminar, 8/05/2013.
- **Max Planck Institute for Mathematics**
Program on Model theory and Applications, 01/06/2012.
- **Imperial College, London**
London Number Theory Seminar, 19/10/2011.
- **The Technion, Israel Institute of Technology, Haifa**
Algebra Seminar, 10/3/2011.
- **The Hebrew University of Jerusalem**
Logic Seminar, 2/2011.
- **Ben Gurion University**
Logic and Set Theory Seminar, 2/2011.
- **Queen Mary College, London**
Anglo-Belgian Workshop on Model Theory and Applications II, 13-14/09/2010.
- **Representation Growth School and Workshop, Sde-Boker, Israel**
Invited speaker, 21-26/6/2009.
- **IHES, Bures-sur-Yvette, France**
Mathematics Seminar, 28/11/2003.
- **University of Paris 7**
Logic Seminar, 28/11/2003
- **University of Lyon**
Logic Seminar, 12/2003.

- **Ben Gurion University, Israel**
Invited speaker at the *Workshop in Honor of Saharon Shelah Winning Wolf Prize and Bolyai Prize*, 20-25/5/2001.
- **Mathematical Sciences Research Institute, Berkeley, California**
Talk at the *Workshop on Model Theory, Algebra, and Arithmetic*, 1/6/1998-5/6/1998.
- **Logic Colloquium 02, Vienna**
Talk, 7/2002.
- **Groups St. Andrews 97 in Bath**
Invited speaker on the Burnside Day, 8/1997.
- **LMS Symposium on Pro-p groups and related areas, Durham**
Invited talk, 14-24/7/1997.
- **University of California, Irvine**
Annual Meeting of the Association for Symbolic Logic, Invited talk, 09/1995.
- **Logic Colloquium 94, Clermont-Ferrand, France**
Talk on 21-30/7/94.

VISITS.

IHES (2018), IPM Tehran (2017, 2018), RIMS Kyoto (2016), ICMS Edinburgh (2014), CUNY New York (2014), Princeton (2014), MSRI, UC Berkeley (2014), UCLA (2014), UC San Diego (2014), Max Planck Institute for Mathematics, Bonn (2012), Ben Gurion University (2011), Technion (2011), Hebrew University (2002-2004, 2011, 2016, 2018), IHES (2002), Lyon/Paris (2002).

COLLEGE EXAMINATIONS.

Setting and marking College Exams at St Hilda's College and St Peter's College every term for 1st and 2nd year undergraduate students in a wide range of courses. These include Linear Algebra (1st and 2nd year), Introduction to University Mathematics, Topology (2nd year), Rings and Modules (2nd year), Number Theory (2nd year), Projective Geometry (2nd year), Manifolds (2nd year), Group Theory (2nd year).

Setting and marking exams for some 3rd and 4th year courses at the Mathematical Institute, University of Oxford.

DEPARTMENTAL EXAMINATIONS.

I wrote and marked the following examinations for the courses I taught.

- Algebraic Curves, 3rd year course, Mathematical Institute University of Oxford, June 2013.
- Model Theory, 4th year course, Mathematical Institute University of Oxford, June 2026.

SUPERVISING UNDERGRADUATE, MASTER'S, AND DOCTORAL THESES.

- Ahmed Ali, D.Phil student, from Oct. 2026 (joint supervision with Prof. Emmanuel Breuillard FRS).
- Flavio Principe, D.Phil student, from Oct. 2026 (joint supervision with Prof. Jochen Koenigsman).
- Flavio Principe, MFOCS Master's Thesis, Hilary term 2026.
- Xinran Liang, *Measures and Integrals on Topological Groups and Zeta Functions*, 3rd year BSP structural dissertation, Mathematical Institute, University of Oxford, 2025-2026.
- Yuchen Fan, I tutored him in several courses at St Hilda's College and advised him in research in number theory. He wrote 3 papers during his undergraduate studies and came **second among all 4th year students in mathematics at Oxford**, won the **Gibbs Prize**, Mathematical Institute, University of Oxford, June 2025.
- John Armitage: I supervised his 4th year Masters MMath Thesis on *Stable Types and Valued Fields*, Mathematical Institute, University of Oxford, Hilary Term 2015 – Trinity Term 2016.

His Thesis was **marked 86/100** by the Department which was **the highest mark** among other MMath Theses that year. He was admitted to do D.Phil with full funding at Oxford University (with Professor Jonathan Pila).

- Mustafa Alper Gunes: I tutored him at St Hilda's College in several Pure Mathematics courses. He became **the top student of the year among all 2nd year maths students at Oxford** in the Final Examinations in July 2021. He got a **Gibbs Prize** from the Department.

He is doing a PhD in Princeton University with Professor Assaf Naor and Professor Peter Sarnak. He has been awarded a Centennial Fellowship by Princeton. **He published 4 papers while he was an undergraduate student.** I discussed with him several research areas of Mathematics every week in addition to the Tutorials during his 3 years of studies at Oxford.

- Tom Wisson: I Tutored in many Pure Maths courses at St Hilda's College. He came **2nd top among all Oxford Maths students in the 3rd year examinations** in 2018. He won a Departmental **Gibbs Prize**.

- Atticus Stonestrom: I Tutored him in several Pure Maths courses at St Hilda's College. He came **9th among all 3rd year Oxford Maths students** in July 2021.

I discussed with him various topics in model theory while he was in his 2nd and 3rd years which led him to **write a paper which he published when he was in his 4th year**.

He is now doing PhD in Notre Dame with full funding.

- Antonin Schrab: I supervised his 3rd Year Undergraduate Thesis on *Integration on locally compact groups and the Riemann Zeta Function after John Tate*. Completed on 3/2017.

The Thesis was marked **80/100 one of the highest in Oxford maths**.

He was admitted to do a PhD at UCL with full funding from UCL. I Tutored him at St Hilda's College in several Pure Maths subjects.

- Yiran Cheng: from *Wellesley College, Boston*, Hilary Term 2015, on tutorials in Logic.
- Joseph Shayani: a 3rd year Mathematics student at Stanford University on the **Stanford-Oxford Program** for excellent students in Hilary Term 2015. I taught him Algebraic Curves.

He came **top in his Stanford Exams** and went on to do. PhD there with visits to various programs in Tel Aviv University.

PASTORAL AND ADMINISTRATIVE WORKS FOR STUDENTS.

- Acting as Personal Tutor to math students (and from joint schools)
- Submitting written reports on each student taught.

ADMINISTRATIVE WORK FOR THE UNIVERSITY OF OXFORD.

As **Dean of Degrees of St Hilda's College**, I organized Matriculation and Graduations of the students of the College in a ceremony in the presence of the Vice-Chancellor of the University of Oxford.

REFEREEING.

- Referee for: Duke Mathematical Journal, Model Theory, Journal de l'école polytechnique – Mathématiques (JEP), Annals of Pure and Applied Logic, Journal of Symbolic Logic, Proceedings of the Edinburgh Mathematical Society, Israel Journal of Mathematics, Communications in Algebra, Archive for Mathematical Logic.

MEMBERSHIP IN PROFESSIONAL SOCIETIES.

- London Mathematical Society
- Association of Symbolic Logic

OUTREACH.

- Gave public talks (together with Prof. Pierrick Bousseau) in St Peter's College, Oxford, to 30 high school students coming to visit the College on **Maths Taster Day** organised by the College, 26/2/2026.

- Gave public talk in St Peter's College to students from the *University of Liverpool Maths School* visiting the College on 15/1/2026.

- Gave public talks in **St Hilda's College 125 Anniversary Drinks Reception, De Morgan House, London Mathematical Society** on *Integer solutions of equations and a Millennium Problem: the Conjecture of Birch and Swinnerton Dyer*, 19/4/2018.

- Gave 4 public talks in the series: In Numbers at *St Mary's University Church*, Oxford, Trinity Term (April and May) 2015, on **Mathematics and Music, Mathematics and Language, Mathematics and Architecture, Mathematics and History**

<https://www.torch.ox.ac.uk/event/maths-and-music>

<https://www.torch.ox.ac.uk/event/maths-and-language>